Assignment 1b, Bulls and Cows

In this assignment you will use the functions that you implemented in part 1a to make a game called Bulls and cows, see https://en.wikipedia.org/wiki/Bulls_and_Cows, where the player plays against the computer.

Clarification of rules:

- The computer generates a secret number. This number contains four different digits in the range 1-9.
- The player tries to guess the number.
- A correct digit in the correct position is a bull.
- A correct digit but in the wrong position is a cow.
- If a digit is counted as a bull, it will not count as a cow.
- The player's guess should be checked so that it is formally correct, i.e., has four digits, does not contain zero and has no repeated digits. These checks should be implemented in the get_player_guess() function
- The program should count how many guesses the player needed and print the result at the end. Invalid guesses ("bad guess") do not count and the last (correct) guess should be included in the count.

After the player has successfully guessed the number (there are four bulls), the computer asks the player if they want to play again. The player chooses by selecting either 0 or 1 (see example further down). The player should also be able to abort the current game by entering -1 instead of a guess and should then be asked if they want to play again.

In this assignment you should use: `int`'s, `bool`'s, `if`, `while` and functions (i.e., no `switch`, no `for`-loops, no arrays, no `string` or other more advanced features)!

Workflow

1. Inspect code and the program outline in the figure further down where an example of functional decomposition is seen.

2. Use functions from part 1a, the only new function you need to implement in this part is a function to get the guess from the player (`get_player_guess()`).

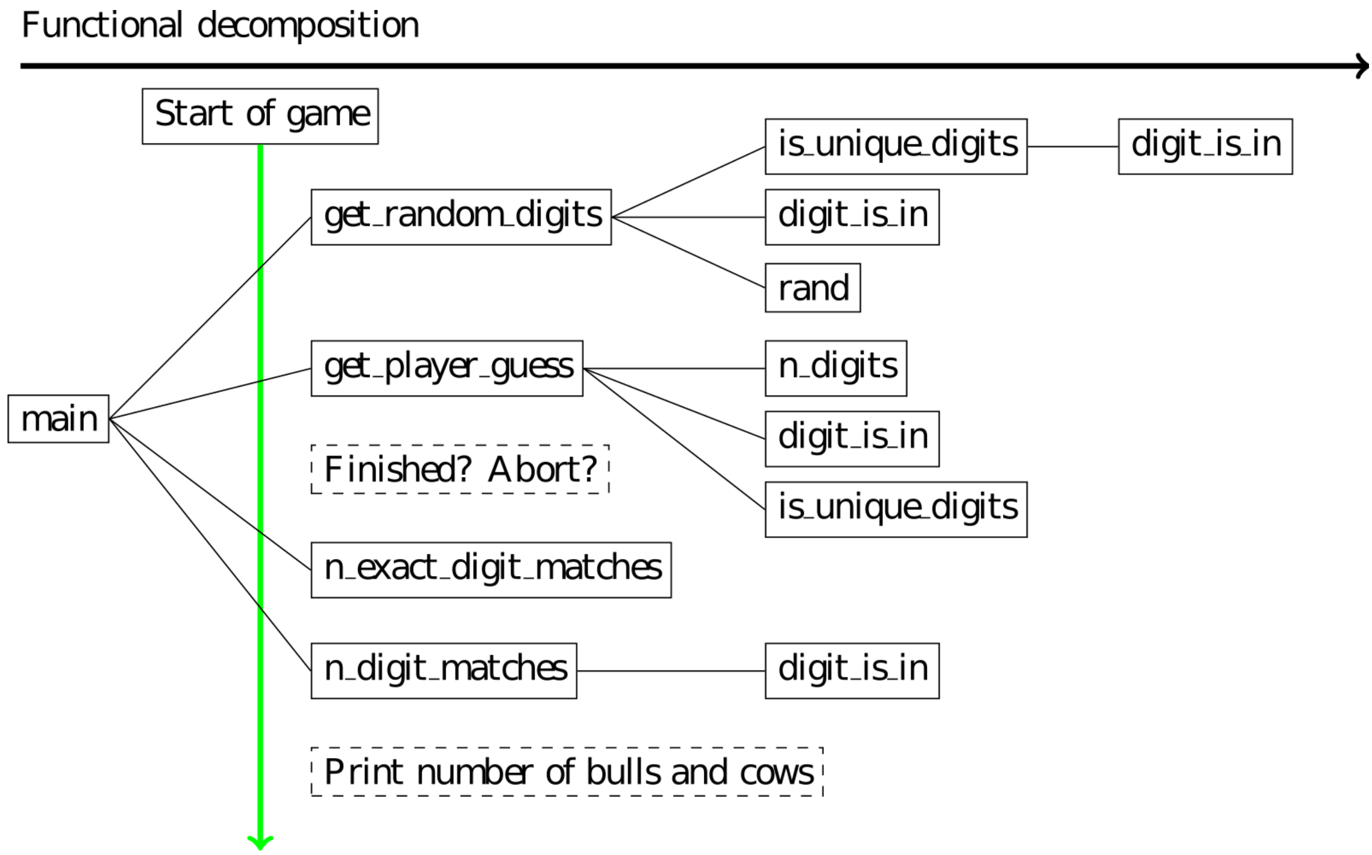
3. Try to make the game work for one single round (implement simplest possible outline).

4. When a single round works, surround with a `while` loop.

Example

Sample run — in these example, the secret numbers to guess are 9721 and 9582, respectively.

```
Welcome to Bulls and Cows
Try to guess a 4 digit number with digits 1-9
and no repeating digits (-1 to abort).

Bulls = correct digit(s) in correct positions.
Cows = correct digit(s).

Guess > 1234
There are 0 bulls and 2 cows
Guess > 12
Bad guess
Guess > 12345
Bad guess
Guess > 1223
Bad guess
Guess > 1230
Bad guess
Guess > -1
Game Aborted

Select 0 or 1 to quit game or to play again: 1

Welcome to Bulls and Cows
Try to guess a 4 digit number with digits 1-9
and no repeating digits (-1 to abort).

Bulls = correct digit(s) in correct positions.
Cows = correct digit(s).

Guess  > 1234
There are 0 bulls and 1 cows
Guess  > 4567
There are 1 bulls and 0 cows
Guess  > 3456
There are 0 bulls and 1 cows
Guess  > 1298
There are 0 bulls and 3 cows
Guess  > 9583
There are 3 bulls and 0 cows
Guess  > 9285
There are 2 bulls and 2 cows
Guess  > 9582
Done, number was 9582 you needed 7 guesses

Select 0 or 1 to quit game or to play again:
```


Functional decomposition
Start of game
main
get_random_digits
is_unique_digits
digit_is_in
digit_is_in
rand
get_player_guess
n_digits
digit_is_in
is_unique_digits
Finished? Abort?
n_exact_digit_matches
n_digit_matches
digit_is_in
Print number of bulls and cows